

NGABO

Mobile Money Intelligence

DEVELOPER ONBOARDING BRIEF

Day-One Technical Handoff Document

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Product	Ngabo — Mobile Money Intelligence
GitHub	github.com/colmeta/kiosk (main)
App ID (target)	com.ngabo.app
App Name	Ngabo
Stack	Capacitor 8.4.1 + Android Java
Current state	Demo prototype — NOT market-ready
Your mission	Strip toys, fix blockers, ship APK
Timeline	~2 weeks focused work

1. What You Are Building

Ngabo is an automated business control system for mobile money agents in Uganda. The app runs silently in the background on an Android phone, intercepts incoming MTN and Airtel transaction SMS messages, and automatically builds a live business ledger — with zero input from the agent.

The core technical loop:

- MTN or Airtel completes a transaction → sends SMS confirmation to agent phone
- Android OS receives the SMS → fires `android.provider.Telephony.SMS_RECEIVED`
- `SmsBroadcastReceiver.java` catches it → extracts sender + body → fires JS custom event
- `www/app.js` Parser module classifies it (deposit/withdrawal/commission) → writes to DB
- Dashboard screen updates in real time — agent never touched the app

Additional features: fraud SMS detection (3-layer), fee + tax calculator, informal cash loan ledger, end-of-day reconciliation audit, multi-kiosk fleet dashboard for owners with multiple locations.

2. Repository Structure

Clone the repo first:

```
git clone https://github.com/colmeta/kiosk.git
cd kiosk
```

File / Folder	
<code>www/index.html</code>	All 9 app screens (963 lines). The UI.
<code>www/app.js</code>	12-module JS engine (2,607 lines). Core logic.
<code>www/styles.css</code>	All styling (3,114 lines). Dark theme.
<code>app-addendum.js</code>	Multi-kiosk dashboard logic. NEEDS FIXING.
<code>capacitor.config.json</code>	App ID: <code>com.ngabo.app</code> . App name: Ngabo.
<code>package.json</code>	Capacitor 8.4.1 dependencies.
<code>android/</code>	Full Android project (Gradle 8.13.0).
<code>android/app/build.gradle</code>	<code>compileSdk 36</code> , <code>minSdk 24</code> , <code>targetSdk 36</code> .
<code>android/app/src/main/java/com/kioskcontrol/app/</code>	Java source — WRONG PACKAGE NAME.
<code>SmsBroadcastReceiver.java</code>	Native SMS interceptor. Correct logic. Wrong package.
<code>MainActivity.java</code>	Extends <code>BridgeActivity</code> . Exposes <code>WebView</code> . Wrong package.
<code>AndroidManifest.xml</code>	SMS permissions declared. Receiver registered.
<code>android/app/src/main/res/values/strings.xml</code>	App name still 'KioskControl'. MUST FIX.

3. The 5 Build Blockers — Fix These First

⚠️The APK will not compile and the app will not work correctly until ALL 5 of these are resolved. Do them in order.

Blocker 1 — Package Name Mismatch (Hard Build Failure)

The capacitor.config.json correctly declares com.ngabo.app. But every Java file and build.gradle still says com.kioskcontrol.app. Android cannot compile a project with a split identity.

Files to change:

- android/app/build.gradle → applicationId and namespace
- android/app/src/main/java/com/kioskcontrol/app/SmsBroadcastReceiver.java → package declaration
- android/app/src/main/java/com/kioskcontrol/app/MainActivity.java → package declaration
- android/app/src/main/res/values/strings.xml → package_name and custom_url_scheme

Also rename the Java source folder itself:

```
mv android/app/src/main/java/com/kioskcontrol/app/  
    android/app/src/main/java/com/ngabo/app/
```

In build.gradle change:

```
namespace = "com.ngabo.app"  
applicationId "com.ngabo.app"
```

In SmsBroadcastReceiver.java and MainActivity.java, first line:

```
package com.ngabo.app;
```

Blocker 2 — App Name Still 'KioskControl' on Device

android/app/src/main/res/values/strings.xml must be updated entirely:

```
<string name="app_name">Ngabo</string>  
<string name="title_activity_main">Ngabo</string>  
<string name="package_name">com.ngabo.app</string>  
<string name="custom_url_scheme">com.ngabo.app</string>
```

Blocker 3 — Hardcoded Fake Kiosk Data (Trust-Destroying Bug)

app-addendum.js has hardcoded locations (Nansana Main, Kyebando Branch, Makindye Branch), hardcoded manager names (Sarah, John, Collin), and hardcoded financial figures. Every new agent who installs Ngabo will see these fake kiosks on their Multi-Kiosk screen.

Fix: Replace the entire static data block in refreshMultiKiosk() with dynamic data from DB.get('kc_credit_profiles') and DB.get('kc_wallets'). If the agent has zero kiosks registered, show an empty state with a [+ Add Kiosk] button. Never hardcode location names or financial figures.

Blocker 4 — Seeder and Simulator Code Must Be Deleted

www/app.js contains Module 11 (Historical Data Seeder) and simulator functions (processSimulatedSMS). www/index.html contains the entire control panel sidebar. These are developer tools that must not exist in production.

Delete from `www/index.html`:

```
<aside class="control-panel" aria-label="Simulator Control Panel">
  ... entire block including all children ...
</aside>
```

Delete from `www/app.js`:

- The entire `seedHistoricalData()` function and Module 12 seeder block
- The `processSimulatedSMS()` function and all simulator pipeline code
- Any call to `seedHistoricalData()` in the `App.init()` or onboarding flow

Also remove the Seed Remote Kiosks button from any remaining HTML.

Blocker 5 — Unencrypted `localStorage` for Financial Data

The DB module currently reads and writes agent financial data to browser `localStorage` which is plain text. This violates Uganda's Data Protection and Privacy Act 2019 and will cause rejection from the Google Play Store Financial App category.

Migration path:

- Install: `npm install @capacitor/preferences`
- Replace `localStorage.setItem(key, value)` with `await Preferences.set({key, value})`
- Replace `localStorage.getItem(key)` with `const { value } = await Preferences.get({key})`
- Since Preferences is async, update `DB.get()` and `DB.set()` to return Promises and update all callers with `await`

Note: Capacitor Preferences uses Android `SharedPreferences` under the hood, which is sandboxed to the app and encrypted on Android 6+. This satisfies both the legal requirement and the Play Store policy.

4. Complete Simulator Removal Checklist

Every item below must be gone before any agent touches the app. These are demo tools, not product features.

Item to remove	
Control panel <aside> block	www/index.html — entire sidebar
'KioskControl Simulator' heading	Inside the aside — will vanish with it
FAKE SMS buttons (3 variants)	Inside the aside — will vanish with it
Seed Remote Kiosks button	Inside the aside — will vanish with it
processSimulatedSMS() function	www/app.js — search and delete
seedHistoricalData() function	www/app.js — Module 12 block
Simulator log panel	Inside the aside — will vanish with it
Any call to seedHistoricalData()	Search app.js for all invocations
Pricing tiers card in Settings	www/index.html — screen-settings section
'Phase 1 MVP' version label	www/index.html — change to 'v1.0'

5. What Is Already Built and Working

These modules are correct and must not be rewritten. Read them carefully before touching anything around them.

Module	
DB module	Complete. localStorage-based. Migrate to Preferences.
Parser module	Complete. MTN + Airtel regex patterns working.
Security module	Complete. 3-layer fraud detection (sender, math, format).
Tariff module	Complete. Tiered fees + 0.5% tax calculator.
KSS engine	Complete. Kiosk Stability Score algorithm.
AIReconciler	Complete. Plain-language shortage diagnostics.
Nav module	Complete. 9-screen router + role-based gating.
Handlers module	Complete. All UI event wiring.
Analytics module	Complete. Canvas bar + area charts.
SmsBroadcastReceiver.java	Logic is correct. Package name wrong. Fix package only.
MainActivity.java	Logic is correct. Package name wrong. Fix package only.
AndroidManifest.xml	SMS permissions declared. Receiver registered correctly.

6. Three Features Still to Build

Feature 1 — Android Notification Channels

The 6 PM 'Money Outside' evening reminder and the float stockout prediction alert are designed in the JavaScript but have no native delivery path. They need Android notification channels registered so they actually fire on-device.

Add to MainActivity.java onCreate():

```
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
    NotificationChannel channel = new NotificationChannel(
        "ngabo_alerts",
        "Ngabo Business Alerts",
        NotificationManager.IMPORTANCE_HIGH
    );
    channel.setDescription("Float warnings and closing reminders");
    NotificationManager nm = getSystemService(NotificationManager.class);
    nm.createNotificationChannel(channel);
}
```

Then expose a Capacitor plugin method that the JS can call to fire a local notification with a given title, body, and delay in milliseconds.

Feature 2 — Ngabo Splash Screen and App Icon

The current splash screen and icons carry KioskControl branding. For Ngabo, the visual identity is:

- Background: #0D1425 (Midnight Navy)
- Logo text: 'NGABO' in gold (#D4AF37), Calibri Bold, centered

Replace android/app/src/main/res/mipmap-* icons at all densities (mdpi, hdpi, xhdpi, xxhdpi, xxxhdpi). Update android/app/src/main/res/drawable/splash.xml or splash.png to use the navy + gold palette.

Feature 3 — First-Time User Walkthrough (3 screens)

When a new agent completes the consent and PIN setup for the first time, show three sequential explanation screens before landing on the dashboard. These are not onboarding forms — they are visual explanations only.

- Screen A: 'Your phone writes your book automatically' — illustration of SMS → ledger flow
- Screen B: 'Log money you lend in 2 taps' — illustration of the Money Outside button
- Screen C: 'We alert you if an SMS looks fake' — illustration of the red fraud overlay

Each screen has a Next button and the last has a 'Start Using Ngabo' button. After seeing all three, set a flag in Preferences so they never show again: `kc_config.walkthroughComplete = true`.

7. Build and Deploy Steps — APK to Physical Device

Complete all 5 blockers first. Then follow this sequence exactly.

Step 1 — Install dependencies

```
cd kiosk  
npm install
```

Step 2 — Sync web assets into Android project

```
npx cap sync android
```

This copies `www/` into `android/app/src/main/assets/public/` — do this every time you change HTML/JS/CSS.

Step 3 — Open in Android Studio

```
npx cap open android
```

Android Studio opens the `android/` folder as a Gradle project. Wait for the Gradle sync to complete (first time takes 3–5 minutes and requires internet for dependency download).

Step 4 — Connect physical device

Enable Developer Options on an Android phone running Android 7+ (preferably 10+):

- Settings → About Phone → tap Build Number 7 times
- Settings → Developer Options → enable USB Debugging
- Connect via USB cable → allow debugging when prompted

The device should appear in the Android Studio device selector dropdown.

Step 5 — Run on device

Click the green Run button in Android Studio (or Shift+F10). Android Studio will build the APK, install it on the connected phone, and launch the app. First build takes 3–5 minutes. Subsequent builds are faster.

Step 6 — Test the SMS bridge with a real SIM

The single most important test in the entire project. With Ngabo installed and running on the phone:

- Insert a real MTN Uganda or Airtel Uganda SIM card
- Complete a mobile money transaction using the standard USSD code (*165# or *185#)
- Watch whether the transaction appears automatically in the Ngabo dashboard

If it does not appear: check Android Studio Logcat, filter by tag 'KioskControl_SMS', and verify the broadcast receiver is firing. Common cause: permission not granted at runtime — check if the permission dialog appeared on first launch.

Step 7 — Generate signed release APK

For distribution outside the Play Store (direct APK install for pilot agents):

- Build → Generate Signed Bundle/APK → APK
- Create a new keystore — save the keystore file and passwords securely
- Select release build variant
- APK output path: `android/app/release/app-release.apk`

This APK can be shared directly via WhatsApp or Google Drive to pilot agents for sideloading.

8. SMS Parser — Pattern Reference

The Parser module in `www/app.js` uses regex to classify incoming messages. These are the current supported patterns. If a real MTN/Airtel message does not match, add the pattern here.

Pattern type	
MTN Cash-In	You have received UGX X from XXXXXXXX. New balance: UGX X. Fee UGX X
MTN Cash-Out	Withdrawn UGX X from your account. New balance: UGX X. Charge UGX X
MTN Commission	Commission of UGX X has been credited. Balance: UGX X
Airtel Deposit	UGX X deposited by XXXXXXXX. Available balance: UGX X. Fee: UGX X
Airtel Withdrawal	UGX X withdrawn. Your new balance is UGX X. Charge: UGX X
Fraud — wrong sender	Message claims MTN origin but sender is a 10-digit number
Fraud — balance mismatch	Stated new balance does not match last known balance +/- amount
Fraud — format fail	Message passes sender check but character structure fails pattern

When testing on a real SIM, capture the exact raw SMS text for each transaction type and verify it matches a pattern. If it does not match, add or adjust the regex. This is the most likely source of bugs in the field.

9. What Not To Do

These decisions are already made. Do not reopen them — every one cost weeks of design work.

- Do not rewrite the UI from scratch. The 9 screens in `index.html` are the correct design. Style changes only.
- Do not add cloud sync, Firebase, or any backend in Phase 1. The app is offline-first by design.
- Do not change the Parser regex patterns without testing against real SMS messages from Ugandan telecoms.
- Do not add new features before the 5 blockers are fixed and the APK builds cleanly.
- Do not remove the consent screen or the PDPO compliance language — this is a legal requirement.
- Do not use 'KioskControl' anywhere in code, strings, comments, or UI. The product is Ngabo.

10. Quick Reference — Day One Checklist

Complete these tasks in order on your first day:

Task	
Clone repo: github.com/colmeta/kiosk	[]
Read this entire brief	[]
Read www/app.js module list (top comments)	[]
Rename package: <code>com.kioskcontrol.app</code> → <code>com.ngabo.app</code>	[]
Update <code>strings.xml</code> : <code>app_name</code> → Ngabo	[]
Delete control-panel <aside> from <code>index.html</code>	[]
Delete seeder + simulator from <code>app.js</code>	[]
Fix hardcoded kiosks in <code>app-addendum.js</code>	[]
Migrate DB module to Capacitor Preferences	[]
Run: <code>npx cap sync android</code>	[]
Open android/ in Android Studio	[]
Gradle sync completes with no errors	[]
Build APK on connected Android device	[]
Test with real MTN/Airtel SIM — SMS fires correctly	[]
Add Android notification channels (6 PM alert)	[]
Replace KioskControl icons with Ngabo navy+gold splash	[]
Build signed release APK for pilot distribution	[]

NGABO

Know your money. Every day.

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github.com/colmeta/kiosk

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